

CS 166: Project Phase 2 Requirements

05/20/2024

Introduction

In this phase you are given an ER-Diagram (a “possible” solution of phase 1) and you are asked to convert it into the ANSI compliant relational model. This ER model has some slight changes from the model given in Phase 1, and it is designed for a simple but generic model for implementation and possible extension on features. So please pay special attention to the given ER model and attributions in this document, instead of sticking to your solution submitted for Phase 1.

The ANSI Relational Model is described in Chapter 3 of your textbook. Please consider any required integrity constraints over relations. In this phase you are not allowed to do any assumptions since you are asked to make a precise translation of the ER model into the relational model. This means you should specify necessary constraint keywords like NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY REFERENCES, etc.

The due date for this phase is **05/20/2024**. Please submit your relational model as a single re-creatable SQL script file containing CREATE TABLE statements with required constraints.

ER-Diagram

The ER-Diagram for our Game Rental application(GameStop) is showed in Figure 1. The attributes of all entities and the attribute domains are listed below.

- Primary keys are shown in bold. Note that your relational tables could have additional attributes due to foreign key constraints.
- The third column below shows attribute-specific constraints; however, this is not an exhaustive list. Additional constraints might be applied due to relationships between tables.

Entity	Attribute Name	Attribute Domain	Additional Constraints
User	login	char(15)	unique, not null
	password	char(50)	not null
	phoneNumber	char(60)	not null
	role	char(10)	not null
	favGames	char(500)	
	numOverdueGames	int	not null
Catalog	gameName	char(50)	unique, not null
	genre	char(40)	not null
	price	float	not null
	description	char(600)	
	imageUrl	char(256)	
RentalOrder	login	char(50)	not null
	orderTimestamp	timestamp	not null
	dueDate	date	not null
	rentalOrderID	char(60)	unique, not null
	totalPrice	float	not null
TrackingInfo	trackingID	char(50)	unique, not null
	status	char(40)	not null
	currentLocation	char(60)	not null
	lastUpdateDate	timestamp	not null
	rentalOrderID	char(50)	not null
	courierName	char(800)	not null
	additionalComments	text	

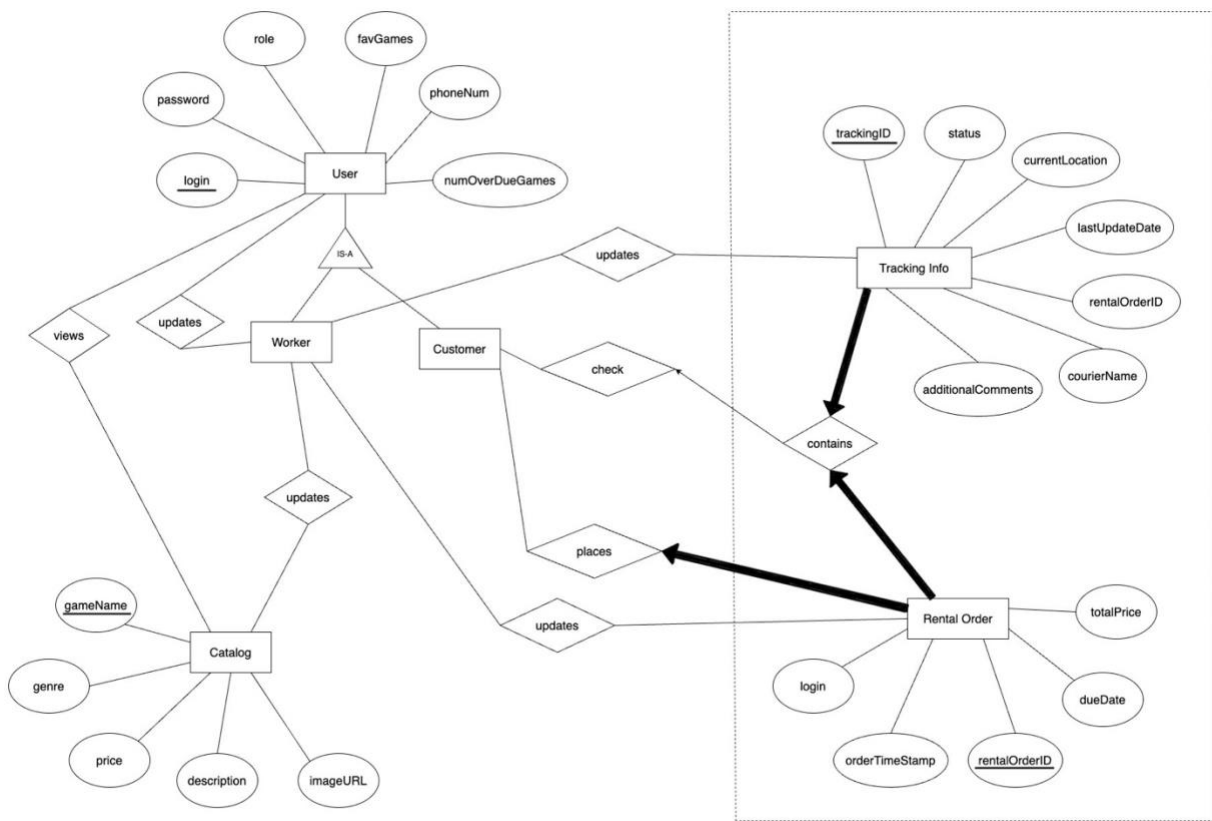


Fig 1 : ER Diagram for GameStop